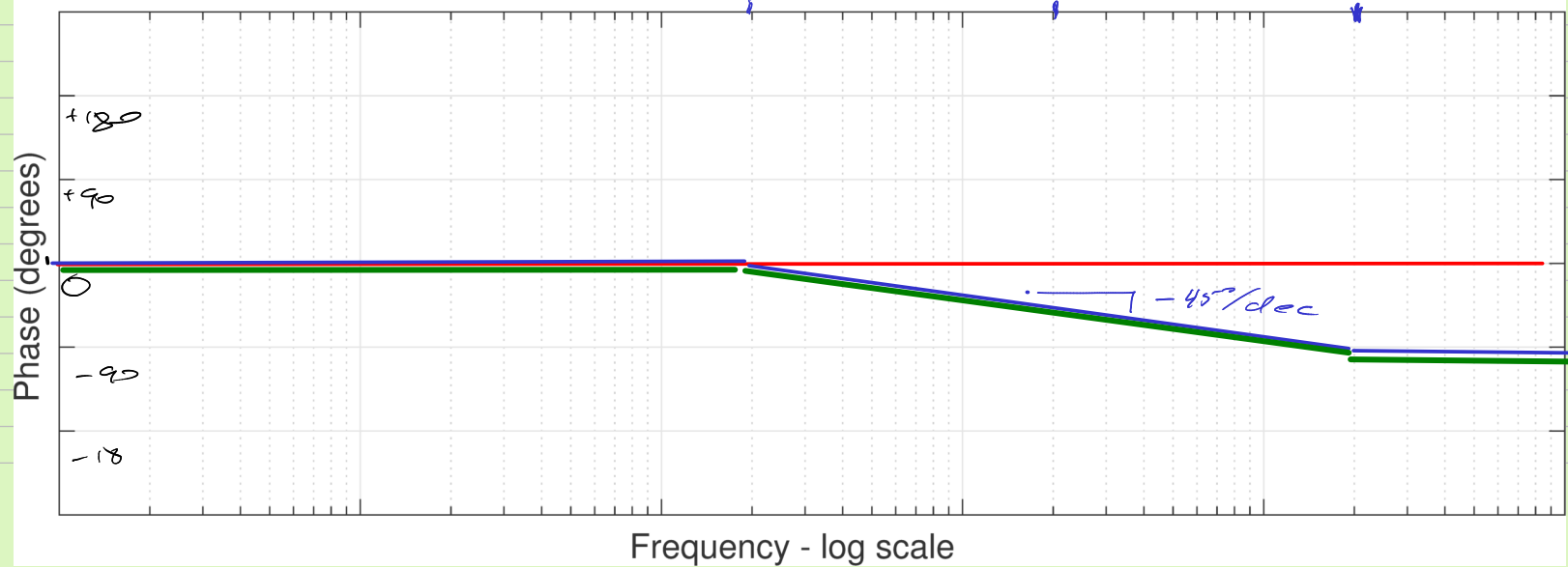
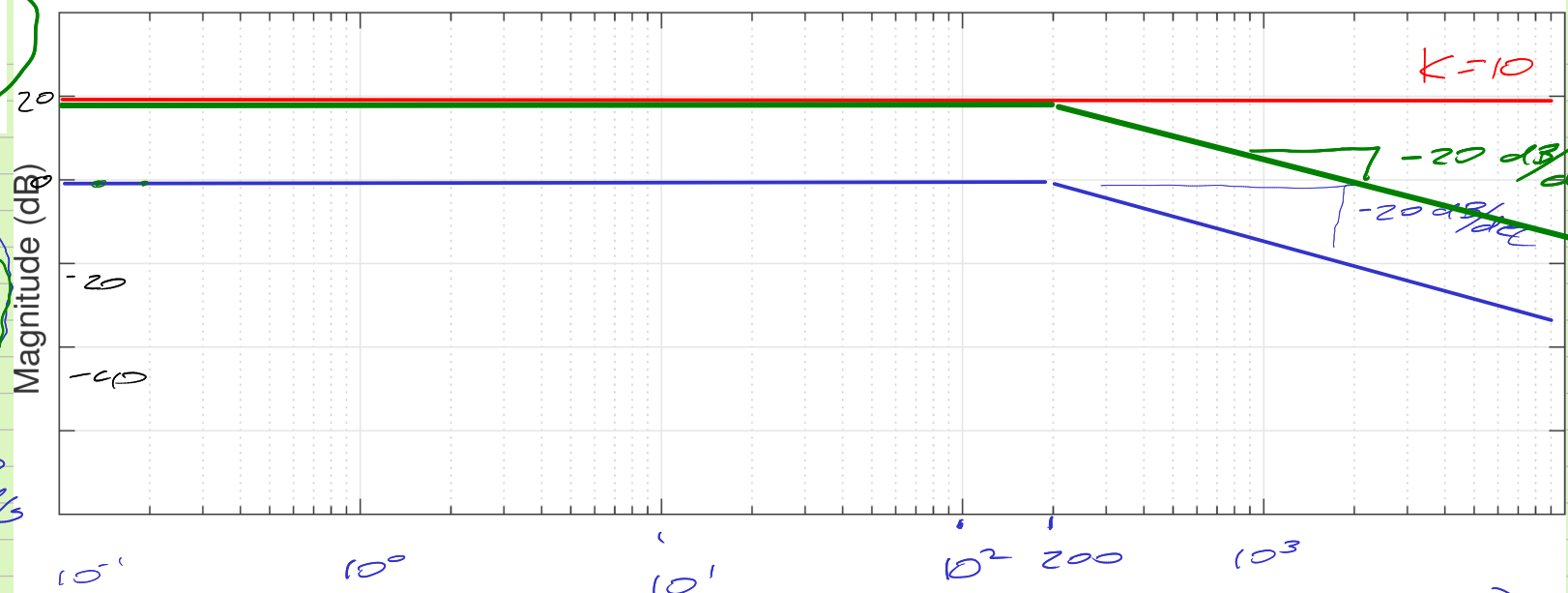


$$L(s) = \frac{2000}{s+200}$$

$$L(s) = \frac{10}{s} \cdot \frac{1}{\frac{s}{200} + 1}$$

$$\omega_c = 200 \text{ rad/s}$$

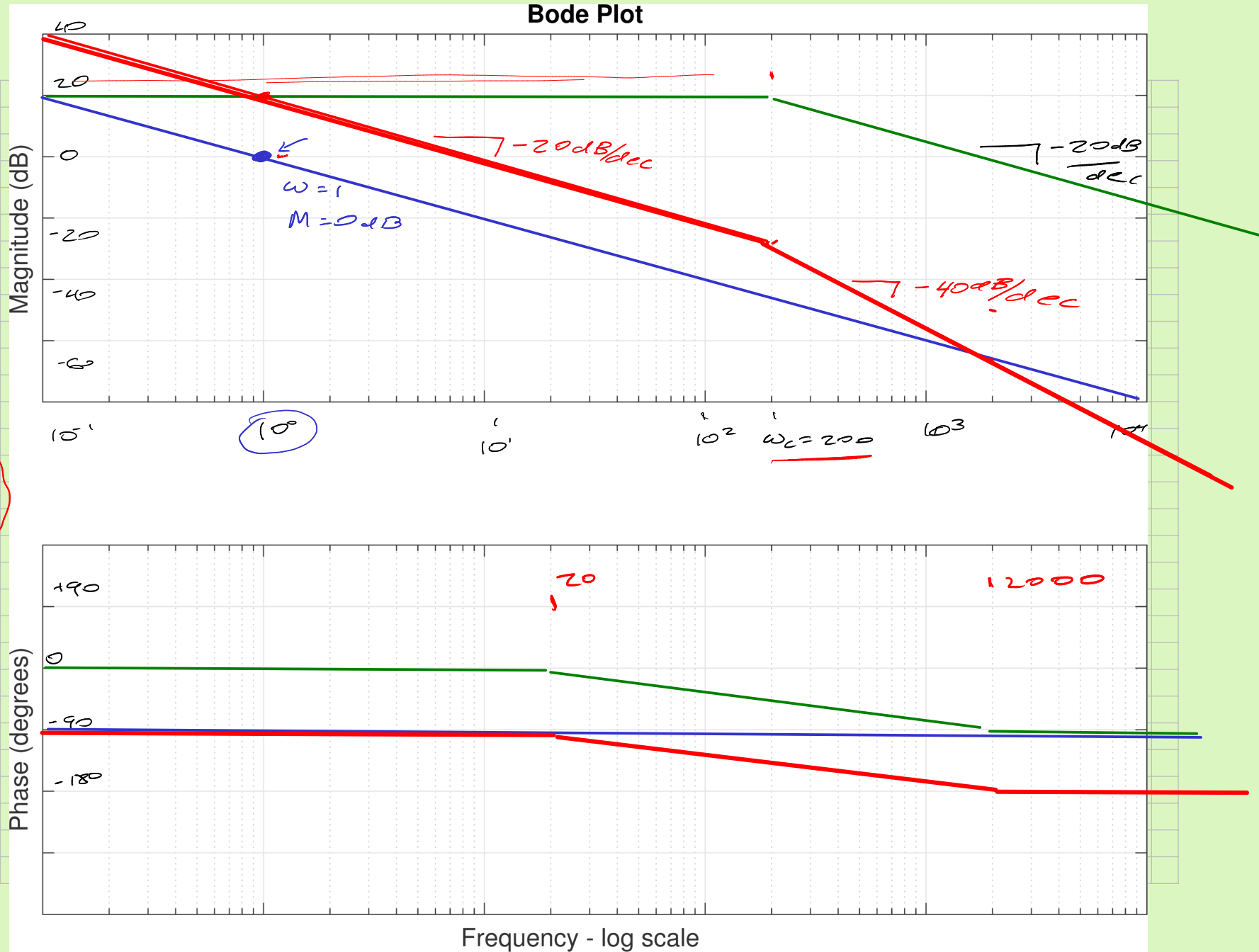
Bode Plot



$$L(s) = \frac{2000}{s+200}$$

$$L(s) = \frac{2000}{s(s+200)}$$

$$L(s) = \left(\frac{1}{s} \right) \left(\frac{2000}{s+200} \right)$$



$$G(s) = \frac{0.1}{(s^2 + s + 4)}$$

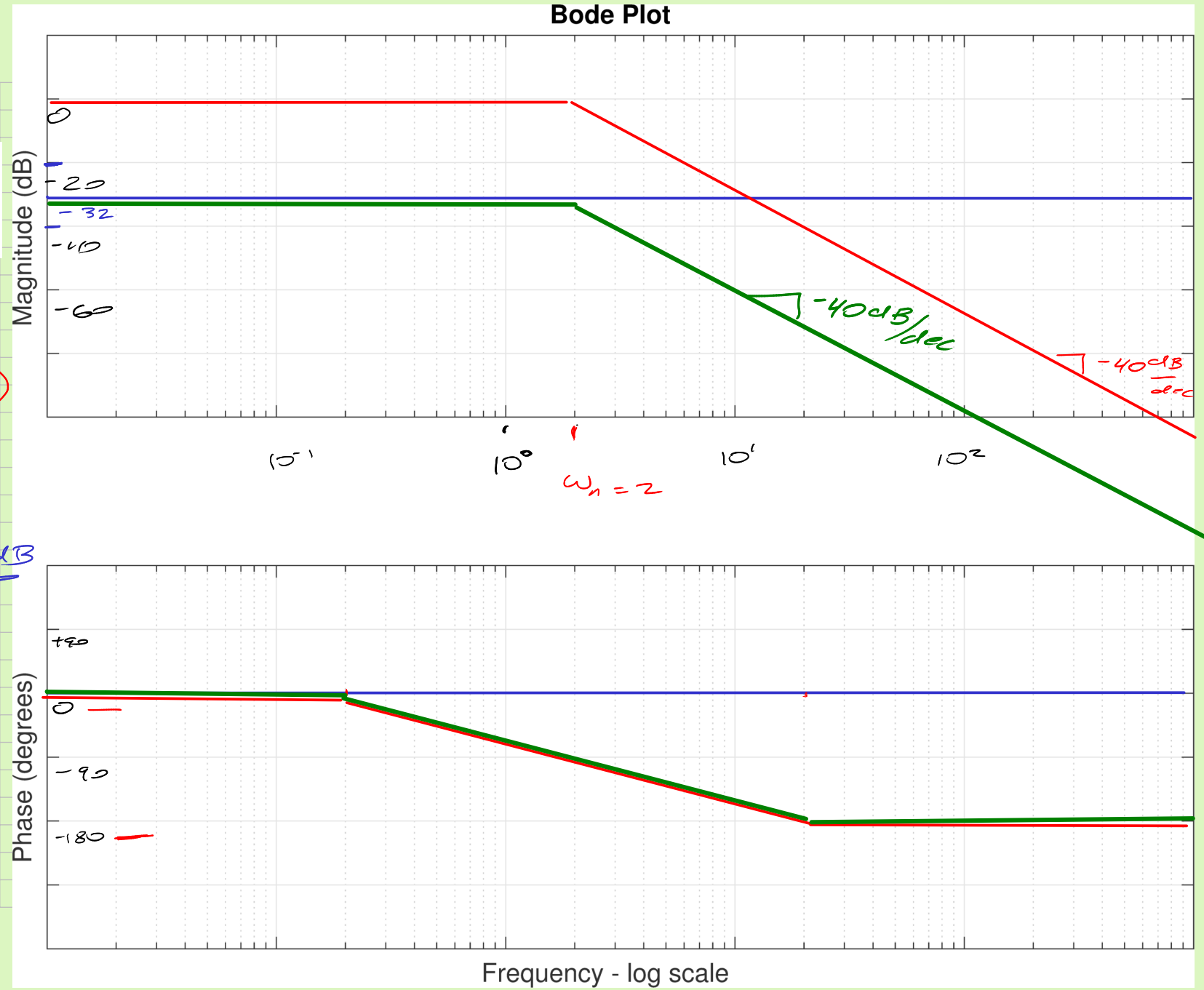
$$G(s) = \frac{1}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}$$

$$G(s) = \frac{0.1}{4} \frac{1}{\left(\frac{s}{2}\right)^2 + \frac{s}{4} + 1}$$

$\omega_n = 2 \text{ rad/s}$

$$20 \log_{10} \left(\frac{0.1}{4} \right) \approx -32 \text{ dB}$$

K



$$G(s) = \frac{0.1(s+10)}{(s^2+s+4)}$$

$G(s) = \frac{(0.1)(10)}{4} \cdot \left(\frac{3}{s} + 1 \right) \cdot \left(\frac{s^2}{4} + \frac{s}{4} + 1 \right)$
 $\downarrow$
 0.25
 $\downarrow$
 -12 dB

